Second Sprint: Literature Review

Due Oct 30 · Sprint review Monday, November 2

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What is a literature review?

- You find out **what others have already done** on your question before you build anything yourself: which algorithms and data structures exist, how people measure them, what results they got.
- **It's not a list of links.** For each source you say what it does, how it does it, and what you take from it for your project.
- **The synthesis is the part that counts:** what is known, which approach you choose and why, and what your project adds.
- **If you use AI**, reconcile your hypothesis ("What may be useful for me?") with the AI's response ("This is useful for you!").

Why now?

- Your research question gets **smaller and sharper** once you know what's been done. Many first-sprint questions were too broad: aim for a **minimum viable product (MVP)** first, stretch goals second.
- You learn **how to measure your result**, for example with a doubling experiment or by counting operations, and what to compare it with.

Lessons from the first sprint

- **A link is not a reference:** complete every entry (author, title, year, source) and stay with one style (APA, MLA, Chicago, or Harvard).
- **Disclose your AI use, always** – even if it's "none used". Your statement and your references must agree.
- **AI-suggested sources that nobody checked:** you must have read every source you cite. A source that does not exist costs points.

What counts as a source?

- **At least five sources, at least two authoritative:** a book, a peer-reviewed article, official documentation, or the documentation of your library.
- **Blogs and videos are fine**, but not as your only sources. AI conversations do not count toward the five.

Where to look

- **The textbook** (Sedgewick and Wayne, *Introduction to Programming in Java*) and its booksite, **Google Scholar, arXiv**, and the official Java documentation.
- **Find a literature review** (a survey) on your topic: someone has already collected and compared the important sources.
- **Litmaps** (`litmaps.com`): start from one paper and see a map of the papers it cites and that cite it.
- **OpenResearch** (`github.com/alphaXiv/OpenResearch`): an open-source workspace for AI research agents that review the literature.
- **Use AI to find candidates and to summarize**, then open and read the source yourself, and verify the citation.

Sprint review

Monday, November 2

in class

- Every team member presents **at least one** of the sources and answers questions about it.
- **Questions?** Ask me early, not at the end of the sprint!